1.Retrieve all data from customers and departments tables

SQLQuery1.s...salini (73))*

```sql
select*from customers
select*from Departments
```

100 % No issues found

Results Messages

	CustomerID	CustomerName	City
1	1	Ravi	Chennai
2	2	Anita	Delhi
3	3	Karan	Mumbai
4	4	Sneha	Bangalore

	DepartmentID	DepartmentName	Location
1	1	HR	Chennai
2	2	IT	Bangalore
3	3	Sales	Mumbai

2.Retrieve all records from employees and orders tables

SQLQuery1.s...salini (73))*

```sql
select*from Employees
select*from orders
```

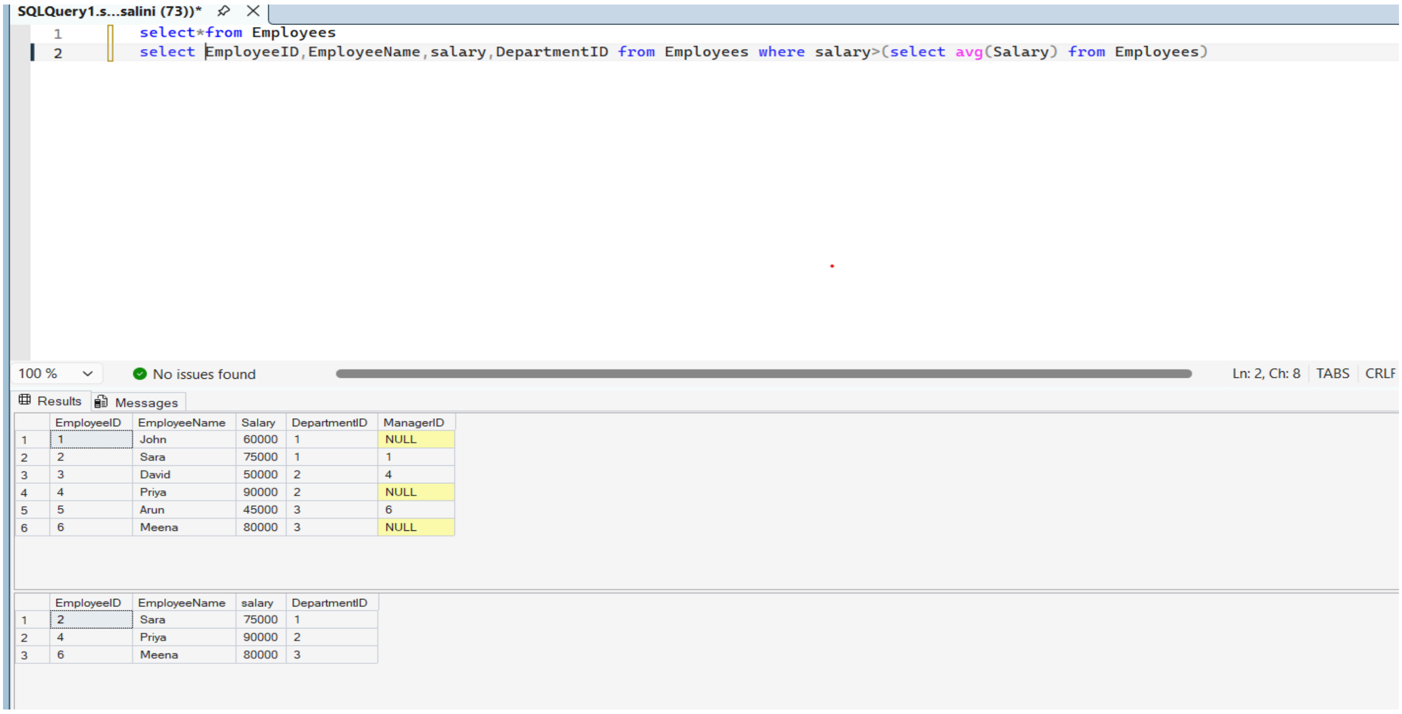
100 % No issues found

Results Messages

	EmployeeID	EmployeeName	Salary	DepartmentID	ManagerID
1	1	John	60000	1	NULL
2	2	Sara	75000	1	1
3	3	David	50000	2	4
4	4	Priya	90000	2	NULL
5	5	Arun	45000	3	6
6	6	Meena	80000	3	NULL

	OrderID	CustomerID	Amount
1	101	1	1000
2	102	2	2000
3	103	1	1500
4	104	3	3000

3.Find employees who earn more than the average salary



SQLQuery1.s...salini (73))*

```sql
select*from Employees
select EmployeeID,EmployeeName,salary,DepartmentID from Employees where salary>(select avg(Salary) from Employees)
```

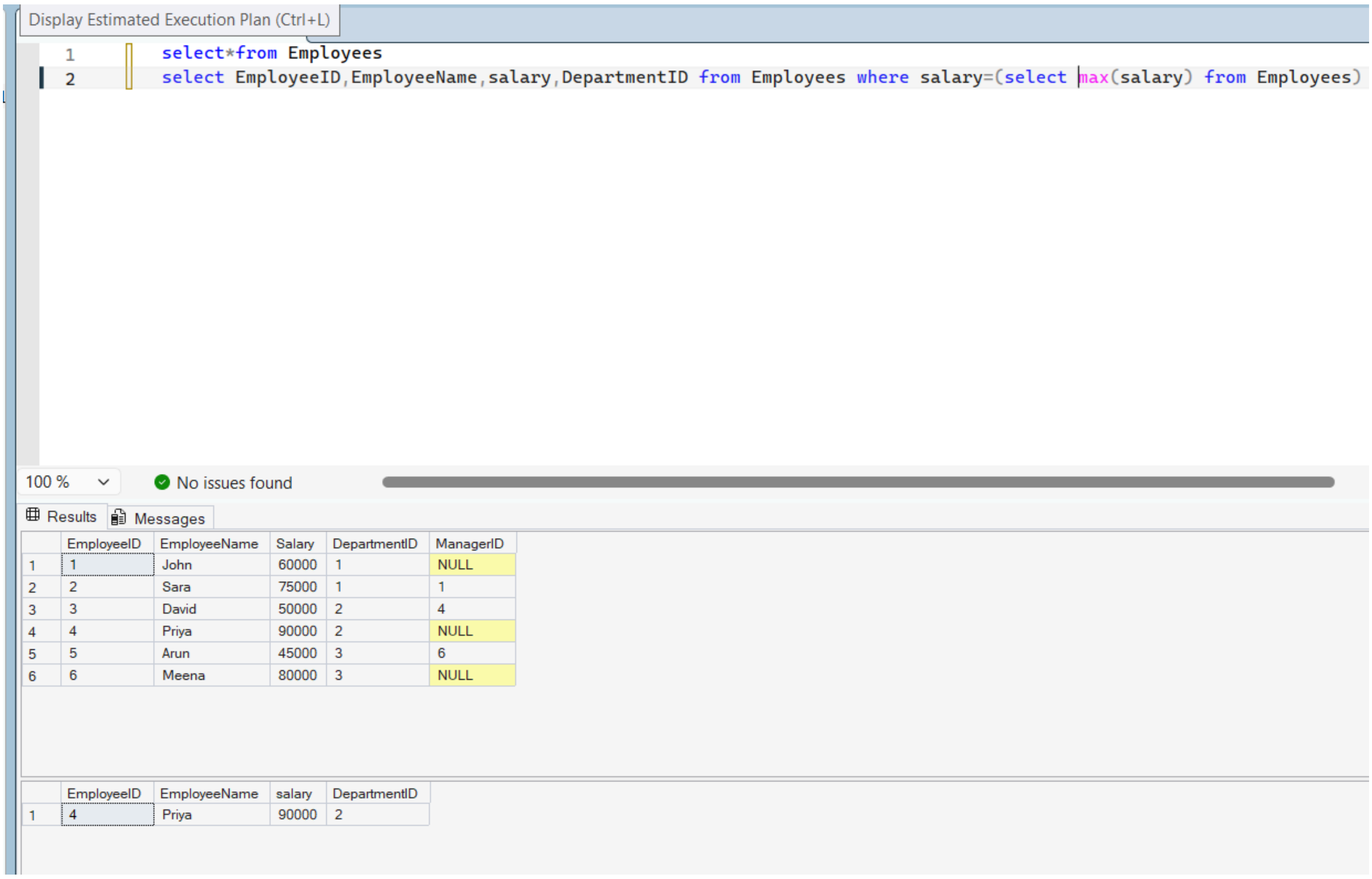
100 % No issues found Ln: 2, Ch: 8 TABS CRLF

Results Messages

	EmployeeID	EmployeeName	Salary	DepartmentID	ManagerID
1	1	John	60000	1	NULL
2	2	Sara	75000	1	1
3	3	David	50000	2	4
4	4	Priya	90000	2	NULL
5	5	Arun	45000	3	6
6	6	Meena	80000	3	NULL

	EmployeeID	EmployeeName	salary	DepartmentID
1	2	Sara	75000	1
2	4	Priya	90000	2
3	6	Meena	80000	3

4.Display the total number of employees in each department



Display Estimated Execution Plan (Ctrl+L)

```sql
select*from Employees
select EmployeeID,EmployeeName,salary,DepartmentID from Employees where salary=(select max(salary) from Employees)
```

100 % No issues found

Results Messages

	EmployeeID	EmployeeName	Salary	DepartmentID	ManagerID
1	1	John	60000	1	NULL
2	2	Sara	75000	1	1
3	3	David	50000	2	4
4	4	Priya	90000	2	NULL
5	5	Arun	45000	3	6
6	6	Meena	80000	3	NULL

	EmployeeID	EmployeeName	salary	DepartmentID
1	4	Priya	90000	2

5.Extract details of all employees working in the IT department

SQLQuery1.s...salini (73))*

```sql
select*from Employees
select
EmployeeID,EmployeeName,Salary,DepartmentID from employees where DepartmentID =
(select DepartmentID from Departments where DepartmentName='IT')
```

110 % No issues found

Results | Messages

	EmployeeID	EmployeeName	Salary	DepartmentID	ManagerID
1	1	John	60000	1	NULL
2	2	Sara	75000	1	1
3	3	David	50000	2	4
4	4	Priya	90000	2	NULL
5	5	Arun	45000	3	6
6	6	Meena	80000	3	NULL

	EmployeeID	EmployeeName	Salary	DepartmentID
1	3	David	50000	2
2	4	Priya	90000	2

6.List employees who earn more than the average salary of the IT department

SQLQuery1.s...salini (73))*

```sql
select*from Employees
select
EmployeeID,EmployeeName,Salary,DepartmentID from Employees where salary>
(select avg(salary) from Employees where DepartmentID=
(select DepartmentID from Departments where DepartmentName='IT'))
```

110 % No issues found

Results | Messages

	EmployeeID	EmployeeName	Salary	DepartmentID	ManagerID
1	1	John	60000	1	NULL
2	2	Sara	75000	1	1
3	3	David	50000	2	4
4	4	Priya	90000	2	NULL
5	5	Arun	45000	3	6
6	6	Meena	80000	3	NULL

	EmployeeID	EmployeeName	Salary	DepartmentID
1	2	Sara	75000	1
2	4	Priya	90000	2
3	6	Meena	80000	3

7.Fetch all customers who have placed at least one order

SQLQuery1.s...salini (73))*

```sql
select*from customers
select customerid,customername,city from customers where customerid in (select customerid from orders)
```

110 %   No issues found

Results   Messages

	CustomerID	CustomerName	City
1	1	Ravi	Chennai
2	2	Anita	Delhi
3	3	Karan	Mumbai
4	4	Sneha	Bangalore

	customerid	customername	city
1	1	Ravi	Chennai
2	2	Anita	Delhi
3	3	Karan	Mumbai

8.Find all customers who have never placed an order

SQLQuery1.s...salini (73))*

```sql
select*from customers
select customerid,customername,city from customers where customerid not in (select customerid from orders)
```

110 %   No issues found   Ln: 2, Ch:

Results   Messages

	CustomerID	CustomerName	City
1	1	Ravi	Chennai
2	2	Anita	Delhi
3	3	Karan	Mumbai
4	4	Sneha	Bangalore

	customerid	customername	city
1	4	Sneha	Bangalore

9.Identify customers who have existing records in the orders table

SQLQuery1.s...salini (73))*

```sql
select*from customers
select
c.customerid,c.customername,c.city from customers c where exists
(select 1 from orders o where o.customerid=c.customerid)
```

110 % No issues found

Results | Messages

	CustomerID	CustomerName	City
1	1	Ravi	Chennai
2	2	Anita	Delhi
3	3	Karan	Mumbai
4	4	Sneha	Bangalore

	customerid	customername	city
1	1	Ravi	Chennai
2	2	Anita	Delhi
3	3	Karan	Mumbai

10.Find customers whose order amount is greater than the average order amount

SQLQuery1.s...salini (73))*

```sql
select*from customers
Select c.CustomerID, c.CustomerName, o.OrderID, o.Amount
from Customers c join Orders o on c.CustomerID = o.CustomerID
where o.Amount > (Select avg(Amount) from Orders)
```

121 % No issues found

Results | Messages

	CustomerID	CustomerName	City
1	1	Ravi	Chennai
2	2	Anita	Delhi
3	3	Karan	Mumbai
4	4	Sneha	Bangalore

	CustomerID	CustomerName	OrderID	Amount
1	2	Anita	102	2000
2	3	Karan	104	3000

11.List employees who earn less than the average salary of the Sales department

SQLQuery1.s...salini (73))*

```sql
select*from Employees
select*from Departments
select employeeid,employeename,salary,departmentid from employees where salary<
(select avg(salary)from employees where departmentid=
(select departmentid from departments where departmentname='sales'))
```

121 % No issues found

Results Messages

	EmployeeID	EmployeeName	Salary	DepartmentID	ManagerID
1	1	John	60000	1	NULL
2	2	Sara	75000	1	1
3	3	David	50000	2	4
4	4	Priya	90000	2	NULL
5	5	Arun	45000	3	6
6	6	Meena	80000	3	NULL

	DepartmentID	DepartmentName	Location
1	1	HR	Chennai
2	2	IT	Bangalore
3	3	Sales	Mumbai

	employeeid	employeename	salary	departmentid
1	1	John	60000	1
2	3	David	50000	2
3	5	Arun	45000	3

12.Find the employee with the second highest salary

SQLQuery1.s...salini (73))*

```sql
select
employeeid,employeename,salary,departmentid from employees where salary=
(select max(salary) from employees where salary<(select max(salary)from employees));
```

110 % No issues found

Results Messages

	employeeid	employeename	salary	departmentid
1	6	Meena	80000	3

13.Find departments whose average salary is greater than the overall average salary

SQLQuery1.s...salini (73))*

```sql
Select d.DepartmentID, d.DepartmentName, avg(e.Salary) as DepartmentAverage
from Departments d
JOIN Employees e on d.DepartmentID = e.DepartmentID
group by d.DepartmentID, d.DepartmentName
having avg(e.Salary) > (Select avg(Salary) from Employees)
```

110 % No issues found

Results Messages

	DepartmentID	DepartmentName	DepartmentAverage
1	1	HR	67500
2	2	IT	70000

14.Find employees who work in departments located in Chennai.

SQLQuery1.s...salini (73))*

```sql
select*from Departments
select*from employees
select employeeid,employeename,salary,departmentid from employees where departmentid in
(select departmentid from Departments where location='chennai')
```

110 % No issues found

Results Messages

	DepartmentID	DepartmentName	Location
1	1	HR	Chennai
2	2	IT	Bangalore
3	3	Sales	Mumbai

	EmployeeID	EmployeeName	Salary	DepartmentID	ManagerID
1	1	John	60000	1	NULL
2	2	Sara	75000	1	1
3	3	David	50000	2	4
4	4	Priya	90000	2	NULL
5	5	Arun	45000	3	6
6	6	Meena	80000	3	NULL

	employeeid	employeename	salary	departmentid
1	1	John	60000	1
2	2	Sara	75000	1

15.Divide all employees into three equal groups based on their salary

SQLQuery1.s...salini (73))*

```sql
select
    employeename,
    salary,
    ntile(3) over (order by salary) as salary_group
from employees;
```

110 % No issues found

Results Messages

	employeename	salary	salary_group
1	Arun	45000	1
2	David	50000	1
3	John	60000	2
4	Sara	75000	2
5	Meena	80000	3
6	Priya	90000	3

16.Rank all employees based on their salary in descending order

SQLQuery1.s...salini (73))*

```sql
select employeename,salary,rank() over (order by salary desc)
as salary_rank from employees;
```

121 % No issues found

Results Messages

	employeename	salary	salary_rank
1	Priya	90000	1
2	Meena	80000	2
3	Sara	75000	3
4	John	60000	4
5	David	50000	5
6	Arun	45000	6

17.Fetch each employee's salary along with the next lower salary

SQLQuery1.s...salini (73))*

```
select employeeid,salary,lead(salary)
over (order by salary desc) as previous_salary from employees
```

121 % No issues found

Results Messages

	employeeid	salary	previous_salary
1	4	90000	80000
2	6	80000	75000
3	2	75000	60000
4	1	60000	50000
5	3	50000	45000
6	5	45000	NULL

18.Find employees with a salary greater than 60000

SQLQuery1.s...salini (73))*

```
select employeename,salary from employees
where salary>60000 order by salary desc
```

121 % No issues found

Results Messages

	employeename	salary
1	Priya	90000
2	Meena	80000
3	Sara	75000

19.List all distinct department locations

SQLQuery1.s...salini (73))*

```
select*from Departments
select distinct location from departments
```

121 % No issues found Ln: 2, Ch:

Results | Messages

	DepartmentID	DepartmentName	Location
1	1	HR	Chennai
2	2	IT	Bangalore
3	3	Sales	Mumbai

	location
1	Bangalore
2	Chennai
3	Mumbai

20.Find employees along with their department names

SQLQuery1.s...salini (73))*

```
select e.employeename,d.departmentname
from employees e
join departments d on d.departmentid=e.departmentid
```

121 % No issues found Ln: 3, Ch: 1

Results | Messages

	employeename	departmentname
1	John	HR
2	Sara	HR
3	David	IT
4	Priya	IT
5	Arun	Sales
6	Meena	Sales

21.Find orders with an amount between 1000 and 2500

```sql
select orderid,amount
from orders where amount between 1000 and 2500
```

	orderid	amount
1	101	1000
2	102	2000
3	103	1500

22.Find the total number of employees in each department

```sql
select departmentid,count(employeeid)as total_emp
from employees group by departmentid
```

	departmentid	total_emp
1	1	2
2	2	2
3	3	2

23.Find the total order amount for each customer

SQLQuery1.s...salini (73))*

```sql
select customerid,sum(amount)as total_amt
from orders group by customerid
```

121 % No issues found Ln: 2, Ch: 19

Results Messages

	customerid	total_amt
1	1	2500
2	2	2000
3	3	3000

24.Find departments that have more than 2 employees

SQLQuery1.s...salini (73))*

```sql
select departmentid,count(employeeid) as count_emp
from employees group by departmentid having count(employeeid)>2
```

121 % No issues found Ln: 2, Ch: 63 TABS CRLF

Results Messages

departmentid	count_emp

25.Find the average salary of each department for employees earning more than 40000

SQLQuery1.s...salini (73))*

```sql
select d.departmentname, avg(e.salary) as total_avg_salary
from departments d
join employees e on d.departmentid = e.departmentid
where e.salary > 40000
group by d.departmentname;
```

121 % No issues found Ln: 5, Ch: 22 TABS CRLF

Results Messages

	departmentname	total_avg_salary
1	HR	67500
2	IT	70000
3	Sales	62500

26.Find the total salary budget of departments in Chennai and Bangalore that have a total budget greater than 100000

SQLQuery1.s...salini (73))*

```sql
select d.departmentname, sum(e.salary) as total_budget
from departments d
join employees e on d.departmentid = e.departmentid
where d.location in ('chennai', 'bangalore')
group by d.departmentid, d.departmentname
having sum(e.salary) > 100000;
```

121 % No issues found Ln: 5, Ch: 38 TABS

Results Messages

	departmentname	total_budget
1	HR	135000
2	IT	140000

27.Display the bonus eligibility status for each employee based on their salary

```sql
select employeename, salary,
 case
 when salary > 60000 then 'eligible for bonus'
 else 'not eligible'
 end as bonus_status
from employees;
```

	employeename	salary	bonus_status
1	John	60000	not eligible
2	Sara	75000	eligible for bonus
3	David	50000	not eligible
4	Priya	90000	eligible for bonus
5	Arun	45000	not eligible
6	Meena	80000	eligible for bonus

28.Display the bonus eligibility status for each employee based on their salary

```sql

create index idx_dept on employees(departmentid)
select * from employees
where departmentid = 2;
```

	EmployeeID	EmployeeName	Salary	DepartmentID	ManagerID
1	3	David	50000	2	4
2	4	Priya	90000	2	NULL

29.Find employees whose names start with the letter 'S'

SQLQuery1.s...salini (73))*

```sql
select employeeid, employeename, salary
from employees
where employeename like 's%';
```

121 % No issues found Ln: 3, Ch

Results Messages

	employeeid	employeename	salary
1	2	Sara	75000

30.Find employees whose names contain the substring 'ar'

SQLQuery1.s...salini (73))*

```sql
select employeename, salary
from employees
where employeename like '%ar%';
```

121 % No issues found Ln: 3, Ch: 15

Results Messages

	employeename	salary
1	Sara	75000
2	Arun	45000